

Fintech: The Digital Evolution of Finance

Fintech is reshaping finance, ushering in a new era of digital prosperity, and changing the way we interact with money.



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Fintech encompasses leveraging and integrating the financial services technology provided by various organisations to offer better customer services. This marks a fundamental shift in the way banking and financial services were previously delivered. It introduces innovative services, embodying a digital transformation philosophy through the adoption of easy and interoperable solutions and services. Initially conceived as backend technology, fintech has evolved to become more customer-centric and now forms the front-end of customer operations. It aims to provide very low-cost transactions and near real-time intelligence in financial transactions.

Key business solutions from fintech

Fintech has brought disruptive changes to the banking and financial services industry, impacting various key areas.

- **Digital payments and mobile banking:** Payments made using mobile apps eliminate the need for explicit interactions with banking systems
- **Voice payments:** Speech-to-text interfaces are being used for making payments
- **Virtual cards:** Payments are being made using virtual credit and debit cards for e-commerce and online transactions, replacing physical cards
- **Decentralised finance (DeFi):** This is being used to

eliminate/minimise service fee and charges

- **Crowdfunding:** Small investments are being solicited from a broad group of investors
- **Lending and other bank-related transactions:** Innovative lending solutions are on offer

Key technologies used by fintech companies

Fintech relies on various financial technologies.

- **Blockchain:** Creating distributed ledgers for enhanced transparency
- **Robotic process automation:** Automating processes to handle repetitive tasks
- **Big Data:** Collecting and processing structured, semi-structured and unstructured data in real time, providing valuable insights
- **Artificial intelligence/machine learning:** Leveraging artificial intelligence and machine learning algorithms for better business results
- **Cloud enablement with software-as-a-service and serverless technologies:** Enabling rapid scalability by leveraging cloud services
- **Quantum computing:** Harnessing the high speed of quantum computers for cybersecurity, fraud detection, risk management, etc
- **Open APIs and open source technologies:** Connecting third-party APIs to financial systems
- **Rapid application development:** Leveraging low-code and no-code tools and techniques

Open source libraries for fintech

Fintech benefits from various open source projects, with some of the popular ones being:

- **Qlib**
 - An open source project from Microsoft
 - A platform for quantitative investment solutions that leverages AI/ML models and market dynamics paradigms
- **FinGPT**
 - A financial LLM (large language model) from AI4 Finance Foundation